INTELLI-CREDIT

AI-Powered Credit Decisioning Engine

Hackathon Pitch Preparation Document

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Team SPB

1. Problem Statement

The Challenge: In Indian corporate lending, credit managers face a 'Data Paradox' - overwhelmed by data yet taking **weeks** to process a single loan application.

Data sources that must be stitched together:

- **Structured:** GST filings, ITRs, Bank Statements
- **Unstructured:** Annual Reports, Financial Statements, Board Minutes, Rating Reports
- **External:** News, MCA filings, Legal disputes (e-Courts)
- **Primary:** Factory site visits, Management interviews

What Was Asked: Build an AI-powered Credit Decisioning Engine that automates end-to-end preparation of a Comprehensive Credit Appraisal Memo (CAM).

Pillar	Requirement	Our Module(s)
1. Data Ingestor	Multi-format parsing, GST vs Bank cross-verification	data_ingestor.py, claude_pdf_processor.py, ai_analyzer.py, cross_ver
2. Research Agent	Web crawling, news/legal/regulatory, qualitative risk flagging	research_agent.py, nlp_engine.py
3. Recommendation Engine	CAM generation, explainable scoring, loan amount & rate	scoring_engine.py, credit_analyzer.py, credit_risk_model.py, cam_ger

Evaluation Criteria:

- **Extraction Accuracy** - How well we extract from messy, scanned Indian PDFs
- **Research Depth** - Finding relevant local news and regulatory filings
- **Explainability** - Can AI walk the judge through its logic
- **Indian Context Sensitivity** - GSTR-2A vs 3B, NBFC nuances, CIBIL

2. Proposed Solution & Methodology

The 8-Step Pipeline

Step	Module	What It Does
1. Data Ingestion	data_ingestor.py	Parses PDFs (PyMuPDF + pdfplumber + OCR), Excel/CSV
2. AI Extraction	claude_pdf_processor.py + ai_analyzer.py	Extracts 30+ financial metrics using Claude + regex
3. Cross-Verification	cross_verifier.py	GST vs Bank reconciliation, circular trading detection
4. Research Agent	research_agent.py	Web crawls for news, legal, MCA filings, regulatory actions
5. NLP Analysis	nlp_engine.py	Sentiment analysis, risk signals, entity recognition
6. ML Prediction	credit_risk_model.py	XGBoost classifier + SHAP explainability
7. Credit Scoring	credit_analyzer.py + scoring_engine.py	Entity classification, 15 ratios, Five Cs scoring
8. CAM Generation	cam_generator.py	Professional Word/PDF Credit Appraisal Memo

Dual-Track Extraction (Unique Approach)

Track A - Claude API: Sends full PDF to Claude 3.5 Sonnet with structured JSON schema for 30+ metrics. Handles complex layouts end-to-end.

Track B - Regex Engine: 10 specialized extraction functions with unit detection, bracket-negative handling. Zero API cost, deterministic, reproducible.

Result: Best of both tracks merged - Claude for accuracy, Regex for reliability.

3. Calculations Performed

3.1 Financial Metric Extraction

Metric	Method
Revenue	Regex: 'Total revenue from operations' from P&L
PAT	Regex: 'Owners of the company' from Consolidated P&L (NOT auditor's report)
Net Worth	Regex: 'Equity attributable to owners' from BS; fallback: Total Assets - Total Debt
Total Debt	Component sum: Debt Securities + Borrowings + Subordinated Liabilities
EBITDA	Direct extraction; NBFCs: PPOP = Revenue - Costs
Interest Expense	From P&L Expenses section: 'Finance costs'
Operating CF	'Net cash from operating activities' with bracket-negative handling
NII, GNPA, NNPA	Percentage extraction for NBFC-specific metrics

3.2 Computed Credit Ratios (15 Ratios)

Ratio	Formula	Benchmark
D/E Ratio	Total Debt / Net Worth	< 2.0
ICR	EBITDA / Interest Expense	> 3.0
DSCR	(PAT + Depr + Interest) / (Interest + Principal)	> 1.5
Current Ratio	Current Assets / Current Liabilities	> 1.5
EBITDA Margin	EBITDA / Revenue x 100	Industry-specific
PAT Margin	PAT / Revenue x 100	Higher is better
Revenue Growth	(Curr - Prior) / Prior x 100	Always algebraic
ROA	PAT / Avg Total Assets x 100	> 2% for NBFCs
ROE	PAT / Avg Net Worth x 100	> 15%
GNPA Ratio	Gross NPA / Advances x 100	< 3%
CRAR/CAR	Capital Adequacy Ratio	> 15% (RBI)

3.3 Cross-Verification Checks

Check	Method
Revenue Reconciliation	GST revenue vs bank credits; flag if > 10% variance
Circular Trading	Matching credit-debit amounts > 1L in bank statements

Check	Method
Revenue Inflation	Coefficient of variation on monthly GST revenue; flag if CV > 0.5
Cash Flow Consistency	Inflows vs outflows ratio analysis
GSTR-2A vs 3B Mismatch	ITC claimed vs auto-populated; flag if > 10% variance

3.4 Five Cs Scoring Model

C	Weight	Key Inputs
Character	25%	Media sentiment, risk signals, legal issues, promoter reputation
Capacity	25%	DSCR, ICR, revenue growth, PAT margin, operating cash flow
Capital	20%	D/E ratio, net worth, total assets, ROE/ROA
Collateral	15%	Current ratio, asset coverage, tangible assets
Conditions	15%	Industry outlook, regulatory risk, sector growth

Final Score = Sum(C_score x Weight) - CV Penalty +/- Qualitative Adjustment

4. AI/ML Models Used & Why

4.1 Models We Use

Model	Purpose	Why This Model
Claude 3.5 Sonnet	PDF extraction	Best document understanding; handles messy Indian PDFs; 200K context
GPT-4o	Narrative/CAM	Superior narrative quality; understands Indian financial context
XGBoost	Risk classification	Industry-standard for tabular data; handles missing values; fast
SHAP	Explainability	Feature-level explanations for every prediction
Custom NLP Lexicon	Sentiment analysis	100+ Indian credit keywords; faster than LLM for batch
Regex Engine	Data extraction	Unit-aware, bracket-negative, multi-column year extraction

4.2 Why NOT These Models

Model	Why Not
FinBERT/BERT	Overkill; our custom lexicon is faster and Indian-context specific
Random Forest	XGBoost consistently outperforms on tabular financial data
LLaMA/Open-source	Needs GPU infrastructure; latency too high; weaker at extraction
Tesseract only	PyMuPDF + pdfplumber handle native PDFs better; Tesseract is fallback

5. How Our Model Is Different

Feature	Traditional Tools	IntelliCredit
Processing Time	2-3 weeks	< 5 minutes
Data Sources	Manual collection	Automated multi-source
Fraud Detection	Post-facto audits	Real-time circular trading + GST mismatch
NBFC Support	One-size-fits-all	Dedicated NBFC pipeline (PPOP, NII, GNPA, CRAR)
Explainability	Black box scores	SHAP + Five Cs + factor-level reasoning
Research	Manual Google	Automated web crawling
CAM Generation	Manual Word doc	Auto-generated professional Word/PDF
Indian Context	Western models	Built for India (Lakhs/Crores, GSTR, NCLT)

Indian Context Sensitivity Features

- **Unit Handling:** Auto-detects Lakhs/Crores/Millions from PDF headers; normalizes to Crores
- **GSTR-2A vs 3B:** Specific cross-verification for ITC mismatch detection
- **NBFC Classification:** Auto-detects from keywords (NII, GNPA, AUM, CRAR); adapts all metrics
- **Indian Accounting:** Handles bracket notation for negatives: (22,489.55) = -22,489.55
- **Credit Ratings:** Extracts CRISIL, ICRA, CARE, India Ratings, Brickwork, Acuite
- **Legal Context:** Monitors NCLT, SFIO, CBI, ED, SEBI, RBI actions
- **10+ Sanity Checks:** PAT < Revenue, GNPA > NNPA, ROA != ROE, CRAR > 10%

6. Implementation Feasibility & Impact

Technical Stack

Layer	Technology
Backend	Python 3.x, Flask
AI/LLM	Claude 3.5 Sonnet (Anthropic), GPT-4o (OpenAI)
ML	XGBoost, SHAP, scikit-learn, NumPy, Pandas
PDF Processing	PyMuPDF (fitz), pdfplumber, Tesseract OCR
NLP	Custom Indian credit lexicon (100+ keywords)
Web Scraping	DuckDuckGo API, BeautifulSoup, Requests
Document Gen	python-docx (Word), ReportLab (PDF)
Frontend	HTML5, JavaScript, Chart.js
Data Integration	Databricks connector ready

Impact Metrics

Metric	Current (Manual)	With IntelliCredit
Processing Time	2-3 weeks	< 5 minutes
Cost per Application	Rs 15,000-25,000	Rs 5-10 (API costs)
Fraud Detection	Post-facto	Real-time
Consistency	Human-dependent	100% standardized
Coverage	8-10 data points	30+ metrics + web research
Explainability	Subjective notes	SHAP + Five Cs + Factors

Code Architecture: 6,500+ lines across 12 core modules

7. Pitch Talking Points

Opening (30 seconds)

> *"Credit managers spend 2-3 weeks per loan application stitching together data from 10+ sources. IntelliCredit does it in under 5 minutes with AI - and catches fraud that humans miss."*

Suggested Demo Flow

1. Upload an NBFC Annual Report PDF - show real-time extraction
2. Point out Net Worth, PAT, Revenue auto-extracted with unit normalization
3. Show Cross-Verification tab - circular trading detection in action
4. Show Research tab - live web-crawled news and legal findings
5. Show Credit Analysis - Five Cs breakdown with SHAP explanation
6. Download CAM - professional 10-section document auto-generated

Key Differentiators to Emphasize

- **Dual-track extraction** (Claude + Regex) - accuracy + reliability
- **NBFC-aware** - not just generic corporate analysis
- **Fraud detection** built into the pipeline (circular trading, GST mismatch)
- **Explainable** - every decision has SHAP + Five Cs + factor-level reasoning
- **Indian context** - Lakhs/Crores, GSTR, NCLT, bracket negatives
- **End-to-end** - from PDF upload to CAM download in one click

Handling Tough Questions

Question	Answer
"How accurate is extraction?"	10+ sanity checks (PAT < Revenue, GNPA > NNPA, etc.). Dual-track extraction provides cross-val
"What about scanned PDFs?"	PyMuPDF primary + OCR fallback for scanned/image-only pages.
"Is the ML model trained on real data?"	Yes, train_from_pdfs() extracts from real annual reports, augments with synthetic data centered on
"How different from CIBIL?"	We analyze unstructured data (annual reports, news, legal) that bureau scores miss. Plus GST-Ba